The residual bound, step by step

figure `fig_kernel_free.png`; script `analysis/kernel_free_bound.py`

Symbols

$\tau,\ \tau_{\mathrm{end}}$	time since onset; window end	x	$= C_2\tau_{\mathrm{end}}$, dimensionless window
y_i	measured rate at τ_i	N	number of samples in the window
C_1, C_2	nominal amplitude $K\dot{M}$ and exponent	$K,\ \dot{M}$	calibrated gain; ramp rate
$W,\ z_{CoM},\ J_P$	weight, CoM height, pivot inertia	Wz_{CoM}	$= J_P C_2^2$ (the identity)
$e_\omega(\tau)$	true rate minus the nominal	$b,\ n$	gyro bias; disturbance
C	fitted baseline (pre-onset median)	r_i	residual $y_i - \hat{\omega}_i$
$\delta t_c,\ \beta$	onset shift of the witness; its sinh coefficient, $\beta = -C_1\sinh(C_2\delta t_c)$	δe_ω	what the shift cannot absorb
$\rho(s)$	deviation forcing (small-angle + GE)	$\bar{\rho}$	window average bound on ρ, eq. (91)
$\dot{\rho}$	its time derivative	$\varphi_{\max}$	10° design box
$\omega_{\max}$	nominal peak rate $K\dot{M}\sinh x$	$l_{\mathrm{arm}},\ \beta_M$	pivot arm; GE coefficient
ΔM_{win}	in-window moment excursion	$E(\tau)$	envelope $\bar{\rho}\sinh(C_2\tau)/(J_P C_2)$, eq. (90)
n_{hi}	residual content above 5 Hz	$\kappa_q,\ s_{\mathrm{med}}$	quiet-window shape ratio; median in-window enhancement (1.60)
$\hat{n}$	noise estimate, Sec. 6		

1 The measurement and the fit

The record in the window and the fitted model are

$$y_i = C_1\big(\cosh(C_2\tau_i) - 1\big) + e_\omega(\tau_i) + b + n_i, \tag{1}$$
$$\hat{\omega}_i = C_1^*\big(\cosh(C_2^*(\tau_i - \delta t_c^*)) - 1\big) + C, \tag{2}$$

and on the pre-onset segment the baseline matches the bias,

$$C \approx b. \tag{3}$$

2 The residual, and what the shift absorbs

Subtracting (2) from (1) with (3),

$$r_i \;=\; y_i - \hat{\omega}_i \;=\; \underbrace{C_1\big(\cosh(C_2\tau_i) - 1\big) + e_\omega(\tau_i)}_{\text{to be absorbed}} \;-\; C_1^*\big(\cosh(C_2^*(\tau_i - \delta t_c^*)) - 1\big) \;+\; n_i. \tag{4}$$

The question is how much of the braced part a shifted cosh can match. Write the true structural part as a shifted branch plus a remainder — this *defines* δe_ω:

$$C_1\big(\cosh(C_2\tau) - 1\big) + e_\omega(\tau) \;=\; C_1\big(\cosh(C_2(\tau - \delta t_c)) - 1\big) + \delta e_\omega(\tau). \tag{5}$$

Expanding the shifted branch with the addition identity,

$$\cosh\big(C_2(\tau - \delta t_c)\big) = \cosh(C_2\delta t_c)\cosh(C_2\tau) - \sinh(C_2\delta t_c)\sinh(C_2\tau), \tag{6}$$

and substituting (6) into (5), the remainder is explicitly

$$\delta e_\omega(\tau) \;=\; e_\omega(\tau) \;-\; \underbrace{\big[-C_1\sinh(C_2\delta t_c)\big]}_{\beta}\sinh(C_2\tau) \;-\; \underbrace{C_1\big(\cosh(C_2\delta t_c) - 1\big)}_{O(\delta t_c^2),\ \text{folded into } C_1^*}\cosh(C_2\tau). \tag{7}$$

So to first order the shift contributes a $\sinh(C_2\tau)$ term with coefficient $\beta = -C_1\sinh(C_2\delta t_c)$, and

$$\delta e_\omega(\tau) \;=\; e_\omega(\tau) - \beta\sinh(C_2\tau). \tag{8}$$

3 Choosing the shift: match the true curve at the window end

One free number, δt_c (equivalently β), and one choice: make the shifted branch meet the true curve at τ_{end}. Setting $\delta e_\omega(\tau_{\text{end}}) = 0$ in (8),

$$\beta \;=\; \frac{e_\omega(\tau_{\text{end}})}{\sinh(C_2\tau_{\text{end}})}, \qquad \delta t_c \;=\; -\frac{1}{C_2}\,\operatorname{arsinh}\Bigl(\frac{\beta}{C_1}\Bigr) \;\approx\; -\frac{e_\omega(\tau_{\text{end}})}{C_1 C_2 \sinh(C_2\tau_{\text{end}})}. \tag{9}$$

Since $e_\omega > 0$ (the true curve rises faster than the nominal), $\delta t_c < 0$: the branch starts *earlier* than the true onset. On the campaign $\beta/C_1 = 0.010$–0.023, so the $O(\delta t_c^2)$ term in (7) is below 0.03% of the amplitude.

Then, because the minimiser in (2) is at least as good as this one particular shifted branch,

$$\boxed{\sum_{i=1}^{N} r_i^2 \;\le\; \sum_{i=1}^{N} \bigl(\delta e_\omega(\tau_i) + n_i\bigr)^2 \quad\Longrightarrow\quad \mathrm{RMS}(r) \;\le\; \mathrm{RMS}(\delta e_\omega) + \mathrm{RMS}(n).} \tag{10}$$

Everything now rests on one question: **how big is δe_ω?**

4 How big is δe_ω: two boundary zeros and one comparison

Step 1 — δe_ω vanishes at both ends. At $\tau = 0$: $e_\omega(0) = 0$ (the deviation starts at rest) and $\sinh(0) = 0$, so $\delta e_\omega(0) = 0$. At $\tau = \tau_{\text{end}}$: zero by the choice (9).

$$\delta e_\omega(0) \;=\; \delta e_\omega(\tau_{\text{end}}) \;=\; 0. \tag{11}$$

Step 2 — the equation δe_ω obeys. The deviation dynamics of the manuscript are $J_P \ddot{e}_\varphi = W z_{CoM}\, e_\varphi + \rho$ with $e_\omega = \dot{e}_\varphi$. Differentiating once and using $W z_{CoM} = J_P C_2^2$,

$$\ddot{e}_\omega \;=\; C_2^2\, e_\omega + \dot{\rho}/J_P. \tag{12}$$

$\sinh(C_2\tau)$ satisfies $(\sinh)'' = C_2^2 \sinh$ — it is a solution of the same equation with zero forcing — so subtracting $\beta \sinh$ changes nothing on the right side:

$$\delta \ddot{e}_\omega \;=\; C_2^2\, \delta e_\omega + \dot{\rho}/J_P, \qquad \delta e_\omega(0) = \delta e_\omega(\tau_{\text{end}}) = 0. \tag{13}$$

Step 3 — comparison (this is the whole proof). Let $\bar{f}(\tau) \ge |\dot{\rho}(\tau)|$ and let w solve the same problem with forcing $-\bar{f}$:

$$\ddot{w} = C_2^2 w - \bar{f}/J_P, \qquad w(0) = w(\tau_{\text{end}}) = 0. \tag{14}$$

Claim: $|\delta e_\omega(\tau)| \le w(\tau)$. Proof in two lines: the difference $u = w - \delta e_\omega$ satisfies $\ddot{u} - C_2^2 u = -(\bar{f} + \dot{\rho})/J_P \le 0$ with $u = 0$ at both ends. If u had a negative interior minimum, there $\ddot{u} \ge 0$ and $-C_2^2 u > 0$, so $\ddot{u} - C_2^2 u > 0$ — a contradiction. Hence $u \ge 0$, i.e. $\delta e_\omega \le w$; the same argument with $u = w + \delta e_\omega$ gives $\delta e_\omega \ge -w$. $\square$

Step 4 — solve (14) in closed form. For constant $\bar{f} = \sup|\dot{\rho}|$, substitution verifies

$$w(\tau) \;=\; \frac{\sup|\dot{\rho}|}{J_P C_2^2} \left(1 - \frac{\cosh\bigl(C_2(\tau - \frac{\tau_{\text{end}}}{2})\bigr)}{\cosh(x/2)}\right) \;\le\; \frac{\sup|\dot{\rho}|}{W z_{CoM}}. \tag{15}$$

(Here $x = C_2\tau_{\text{end}}$: at $\tau = 0$ and $\tau = \tau_{\text{end}}$ the numerator equals $\cosh(x/2)$, so the bracket vanishes at both ends; it peaks at mid-window, where the bound is $(1 - \operatorname{sech}(x/2)) \sup|\dot{\rho}|/W z_{CoM}$.) **No exponential appears.** The envelope (90) grows like $\sinh(C_2\tau)$ because nothing stops the unstable mode of (12); here the two boundary zeros of (11) — one free, one bought with δt_c — pin that mode down. This is what "the fit absorbs the growing part" means as an equation.

Step 5 — the forcing rate, from the same channels as $\bar{\rho}$. Differentiating each remainder and bounding on the 10° box, with $\omega_{\max} = K\dot{M} \sinh x$:

$$\sup|\dot{\rho}| \;\le\; \underbrace{\Bigl(W l_{\text{arm}} \varphi_{\max} + W z_{CoM} \tfrac{\varphi_{\max}^2}{2}\Bigr)\, \omega_{\max}}_{\dot{\rho}_2,\ \text{decays like } e^{2C_2(s-\tau_{\text{end}})}} \;+\; \underbrace{|\beta_M|\bigl(\dot{M}\varphi_{\max} + \Delta M_{\text{win}}\, \omega_{\max}\bigr)}_{\dot{\rho}_1,\ \text{decays like } e^{C_2(s-\tau_{\text{end}})}}. \tag{16}$$

Every constant is available before the experiment: W, z_{CoM} from the scale, J_P from CAD, l_{arm} from geometry, $\varphi_{\max}$ the design box, β_M from the GE model, $\dot{M}$ and the window from the protocol.

Step 6 (optional tightening, factor 8) — use the decay of $\dot\rho$. The forcing is not constant: it peaks at the window end and decays backwards as marked in (16) (worst measured violation of these shapes 2.6%; a 1.05 factor is carried on $\dot\rho_2$). So solve (14) once for each shape at unit peak — the same two-line comparison applies, and the solutions are still elementary. In the scaled variable $u = C_2(\tau - \tau_{\text{end}}) \in [-x, 0]$, for the forcing $e^{2C_2(\tau-\tau_{\text{end}})} = e^{2u}$:

$$w_2(\tau) \;=\; \frac{\dot\rho_2}{J_P C_2^2}\cdot\frac{1}{3}\left[\frac{\sinh(u+x)}{\sinh x} + e^{-2x}\,\frac{\sinh(-u)}{\sinh x} - e^{2u}\right], \qquad M_2(x) \;\triangleq\; \max_{u\in[-x,0]} \frac{1}{3}\big[\,\cdots\big], \tag{17a}$$

and for the forcing $e^{C_2(\tau-\tau_{\text{end}})} = e^{u}$, which is *resonant* — it decays at the operator's own rate, so the particular solution carries a factor of u:

$$w_1(\tau) \;=\; \frac{\dot\rho_1}{J_P C_2^2}\left[\frac{x}{2}\,\frac{\sinh(u+x)}{\sinh x} - \frac{(u+x)\,e^{u}}{2}\right], \qquad M_1(x) \;\triangleq\; \max_{u\in[-x,0]} \big[\,\cdots\big]. \tag{17b}$$

Both are verified by substitution: the exponential is the particular solution (e^{2u} with coefficient $1/(2^2-1) = 1/3$; e^u resonant), the sinh terms are homogeneous, and both brackets vanish at $u = -x$ and $u = 0$ — the two boundary zeros of (14). Their maxima barely depend on x and tend to simple constants,

$$M_2(x) \to \tfrac{1}{12}, \qquad M_1(x) \to \tfrac{1}{2e}, \qquad \text{so} \qquad |\delta e_\omega| \;\le\; \frac{M_2\dot\rho_2 + M_1\dot\rho_1}{W z_{CoM}}. \tag{17}$$

(Over the campaign range $x = 1.9$–3.7: $M_2 = 0.074$–0.083, $M_1 = 0.141$–0.181; the script evaluates (17a)–(17b) exactly.) The interpretation is the same as the manuscript's reduction factors 1/7 and 1/5: the forcing is largest exactly where the pinned response is smallest — at the window end the boundary zero sits right under the peak of the load.

5 The pre-onset segment

Where the segment comes from. By (9), $\delta t_c < 0$: the witness branch starts $|\delta t_c|$ *before* the true onset. On the segment $\tau \in [\delta t_c, 0]$ — length $|\delta t_c|$ — the record is still at its baseline, but the branch has already been running for $u = \tau - \delta t_c \in [0, |\delta t_c|]$ since its own onset, so the mismatch there is the branch itself:

$$\text{mismatch}(u) \;=\; C_1\big(\cosh(C_2 u) - 1\big), \qquad u \in [0,\, |\delta t_c|]. \tag{18a}$$

The residual sum runs over this segment too, so the witness's cost in (10) gains its square. Normalising by the main window length τ_{end} (conservative: the extended window is longer, so dividing by the shorter τ_{end} can only enlarge the term),

$$\Delta^2_{\text{pre}} \;=\; \frac{1}{\tau_{\text{end}}}\int_0^{|\delta t_c|} \big[C_1\big(\cosh(C_2u) - 1\big)\big]^2\,du. \tag{18b}$$

The segment's *noise* adds no separate term: $\hat n$ is an RMS density, unchanged by a slightly longer window.

The integral is elementary. Expand the square with $\cosh^2 a = \frac{1}{2}(1 + \cosh 2a)$:

$$\big(\cosh a - 1\big)^2 = \cosh^2 a - 2\cosh a + 1 = \tfrac{3}{2} + \tfrac{1}{2}\cosh(2a) - 2\cosh a, \tag{18c}$$

and integrate term by term with $a = C_2 u$ ($\frac{3}{2} \to \frac{3u}{2}$; $\frac{1}{2}\cosh(2C_2u) \to \sinh(2C_2u)/(4C_2)$; $-2\cosh(C_2u) \to -2\sinh(C_2u)/C_2$):

$$\Delta^2_{\text{pre}} \;=\; \frac{C_1^2}{\tau_{\text{end}}}\left[\frac{3u}{2} + \frac{\sinh(2C_2u)}{4C_2} - \frac{2\sinh(C_2u)}{C_2}\right]_0^{|\delta t_c|}. \tag{18}$$

Check, and why the term is small. Expanding the bracket in powers of u, the u and u^3 orders cancel identically and the first surviving order is

$$\int_0^{|\delta t_c|} \big(\cosh(C_2u) - 1\big)^2 du \;\approx\; \frac{C_2^4\,|\delta t_c|^5}{20},$$

fifth order in the shift, because the branch leaves the baseline only quadratically ($\cosh -1 \approx C_2^2u^2/2$, squared $\to u^4$). Bounding instead by (segment sup)×(length) would give $|\delta t_c|^5/4$ — the exact integral is a factor $\sqrt{5} \approx 2.2$ tighter. Evaluated with the a priori $|\delta t_c|$ from (9) and $e_\omega(\tau_{\text{end}}) \le E(\tau_{\text{end}})$: $\Delta_{\text{pre}} = 0.025°/\text{s}$ at $\dot M = 0.10$ and under 0.005 above 0.30 N m/s.

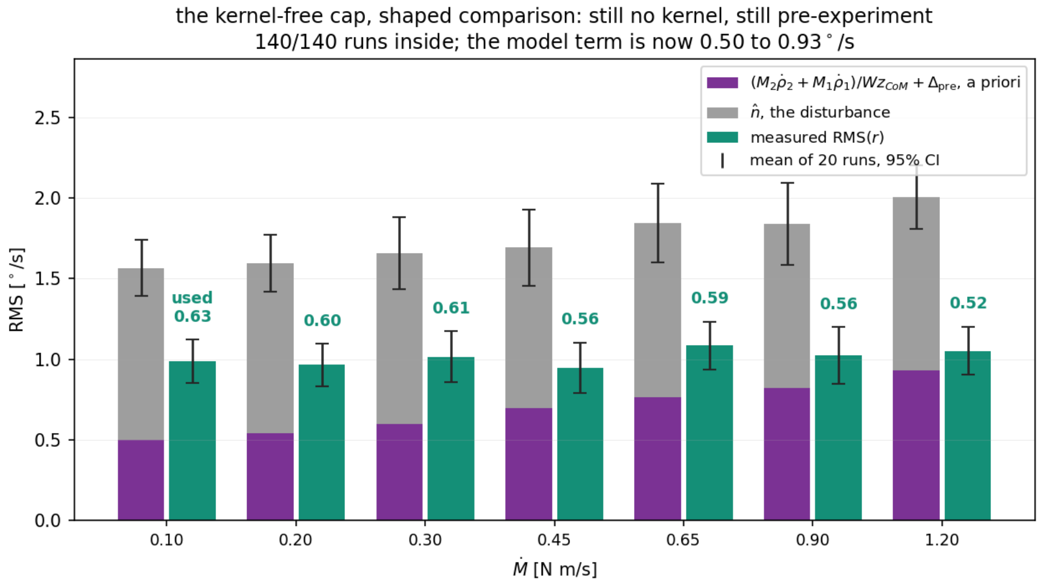

Figure 1: Equation (20) against the campaign: means over the twenty runs at each rate, 95% t intervals. Purple is the model term of (17)+(18), grey the noise estimate (19), green the measured residual. All 140 runs are inside; the used fraction is 0.52–0.63 and the cap is flat, like the residual.

6 The noise term

Above 5 Hz the model can produce nothing, so n_{hi} is measured from the record; the full band is estimated through the quiet-window shape ratio,

$$\hat n \;=\; \mathrm{RMS}(n_{hi})\sqrt{1 + (s_{med}\,\kappa_q)^2}, \tag{19}$$

with $s_{med} = 1.60$ the campaign-median in-window enhancement and κ_q the run's own quiet ratio (campaign median 0.37 where no quiet stretch exists).

7 The check on all 140 runs

$$\boxed{\mathrm{RMS}(r) \;\le\; \underbrace{\frac{M_2\dot\rho_2 + M_1\dot\rho_1}{W z_{CoM}} + \Delta_{pre}}_{\text{model, before the experiment}} \;+\hat n} \tag{20}$$

$\dot M$ N m/s	model (17)+(18) °/s	of it (18) °/s	$\hat n$ °/s	cap (20) °/s	measured °/s	used
0.10	0.501	0.025	1.064	1.56	0.987	0.63
0.20	0.543	0.010	1.050	1.59	0.964	0.60
0.30	0.599	0.006	1.059	1.66	1.014	0.61
0.45	0.698	0.004	0.993	1.69	0.947	0.56
0.65	0.764	0.002	1.080	1.84	1.084	0.59
0.90	0.819	0.001	1.020	1.84	1.023	0.56
1.20	0.932	0.001	1.073	2.00	1.052	0.52

Inside on 140/140 runs; per-run used fraction p10 0.44, median 0.59, worst 0.84. On the exact nonlinear solution the realised δe_ω is 0.16–0.17°/s; the remaining factor ∼3 of (17) over it is the 10° design box against the 4.8° the runs actually reach — the price of stating the bound before flying.

8 The two answers this gives

Why the residual is flat in $\dot{M}$. Nothing in (17) grows with the window — M_2, M_1 are constants and the channel sups barely move — so the theory *predicts* a rate-flat residual, and the data agrees: the measured $\mathrm{RMS}(r)$ stays at 0.95–1.08°/s across a twelvefold change in $\dot{M}$ (correlation with $\dot{M}$: $r = +0.09$, $p = 0.31$). The envelope (90), which grows like $\sinh x$, bounds e_ω — not the residual — and comparing the residual against it is what made the old cap look loose.

How far the shift moves the critical moment. The absorbed shift is (9), and the moment error it causes cancels $\dot{M}$ exactly:

$$\left|\Delta M_{\mathrm{crit}}\right| = \dot{M}\,|\delta t_c| \approx \dot{M}\cdot\frac{e_\omega(\tau_{\mathrm{end}})}{C_1C_2\sinh x} = \frac{W z_{CoM}\,e_\omega(\tau_{\mathrm{end}})}{C_2\sinh x} \leq \frac{W z_{CoM}\,E(\tau_{\mathrm{end}})}{C_2\sinh x} = \bar{\rho}, \tag{21}$$

using $\dot{M}/C_1 = W z_{CoM}$ and $E(\tau_{\mathrm{end}}) = \bar{\rho}\sinh x/(J_P C_2)$. So the shift can displace the identified threshold by at most the window-averaged remainder — the same $\bar{\rho}$ the manuscript already budgets — and measured it uses 0.36–1.85 mN m of the ≈ 12 allowed, the sign lowering M_{crit} (earlier onset).

Remarks

(a) Without the differential equation. The same δe_ω can be written directly from the manuscript's Duhamel form (89), giving $\delta e_\omega = \frac{1}{J_P}\int_0^{\tau_{\mathrm{end}}} K(\tau,s)\rho(s)\,ds$ with a kernel that the addition identities collapse to $K = \sinh(C_2(\tau_{\mathrm{end}}-\tau))\cosh(C_2 s)/\sinh(C_2\tau_{\mathrm{end}})$ for $s \leq \tau$ (minus its mirror for $s > \tau$), and $\max|K| = 1$ exactly — the exponential cancels inside the kernel. It is an instructive identity, but as a bound it is weaker: K changes sign at $s = \tau$, so only $\int |K|$ can be used, and that costs a factor ≈ 2.4 over (17). The derivative bound in (16) is what forbids ρ from flipping sign with K, and that smoothness is worth exactly that factor.

(b) What is a priori and what is not. The model term of (20) uses only pre-experiment constants (list under (16)). The noise term (19) is measured: n_{hi} from the run itself, κ_q from quiet stretches, s_{med} one campaign median. A fully pre-experiment cap replaces $\hat{n}$ with a bench noise figure.

(c) Exponent mismatch. If the true exponent differs from the pinned C_2 by a relative ε_2, the mismatch joins e_ω by definition and its unabsorbed part costs 0.018–0.022°/s per 1% of ε_2 — 5–7% of the model term, and the calibration residuals limit ε_2 to a few percent.